第五周作业答案

1 非齐次边界条件的热传导方程

$$\begin{cases} \dfrac{\partial u}{\partial t} - \kappa \dfrac{\partial^2 u}{\partial x^2} = 0, \\ u|_{x=0} = Ae^{-\alpha^2 \kappa t},\ u|_{x=l} = Be^{-\beta^2 \kappa t}, \\ u|_{t=0} = 0 \end{cases}$$

解法一（展开法）

对$u(x,t)$用本征函数组$X_n(x) = \sin \frac{n\pi x}{l}$展开，此时$\lambda_n = (\frac{n\pi}{l})^2$：

$$u(x,t) = \sum_{n=1}^{\infty} T_n \sin \frac{n\pi x}{l}$$

对方程左右两边积分：

$$\frac{2}{l}\int_0^l (u_t - \kappa u_{xx}) X_n dx = 0$$

$$= \frac{2}{l}\frac{\partial}{\partial t}(\int_0^l u X_n dx) - \frac{2}{l}\kappa \int_0^l u_{xx} X_n dx$$

$$= T_n' + \kappa \lambda_n T_n - \kappa \frac{2}{l}(u_x X_n - u X_n')|_0^l$$

可得

$$T_n' + \kappa \lambda_n T_n = \frac{2\kappa}{l}(-\frac{nB\pi(-1)^n}{l}e^{-\beta^2 \kappa t} + \frac{nA\pi}{l}e^{-\alpha^2 \kappa t})$$

根据齐次化原理或常数变易法可得：

$$T_n(t) = \int_0^t \frac{2\kappa}{l}(-\frac{nB\pi(-1)^n}{l}e^{-\beta^2 \kappa \tau} + \frac{nA\pi}{l}e^{-\alpha^2 \kappa \tau})e^{\kappa \lambda_n (t-\tau)} d\tau$$

$$= [-\frac{2nB\pi(-1)^n}{(n\pi)^2-(\beta l)^2}(e^{((\frac{n\pi}{l})^2-\beta^2)\kappa t}-1)+\frac{2nA\pi}{(n\pi)^2-(\alpha l)^2}(e^{((\frac{n\pi}{l})^2-\alpha^2)\kappa t}-1)]e^{\kappa\lambda_n t}$$

最后可得：

$$u(x,t)=\sum_{n=1}^{\infty}(-\frac{2nB\pi(-1)^n}{(n\pi)^2-(\beta l)^2}(e^{-\beta^2\kappa t}-e^{-\kappa\lambda_n t})+\frac{2nA\pi}{(n\pi)^2-(\alpha l)^2}(e^{-\alpha^2\kappa t}-e^{-\kappa\lambda_n t}))\sin\frac{n\pi x}{l}$$

解法二（边界条件齐次化）

设

$$u(x,t)=v(x,t)+w(x,t)$$

其中设$v(x,t)$形式为：

$$v(x,t)=A(1-\frac{x}{l})e^{-\alpha^2\kappa t}+B\frac{x}{l}e^{-\beta^2\kappa t}$$

则$w(x,t)$满足方程

$$\begin{cases}\dfrac{\partial w}{\partial t}-\kappa\dfrac{\partial^2 w}{\partial x^2}=A(1-\dfrac{x}{l})\alpha^2\kappa e^{-\alpha^2\kappa t}+B\dfrac{x}{l}\beta^2\kappa e^{-\beta^2\kappa t},\\ w|_{x=0}=w|_{x=l}=0,\\ w|_{t=0}=-A(1-\dfrac{x}{l})-B\dfrac{x}{l}\end{cases}$$

对$w(x,t)$进行本征函数展开

$$w(x,t)=\sum_{n=1}^{\infty}T_n(t)\sin\frac{n\pi}{l}x$$

对原方程两侧同时进行傅里叶展开得到的系数方程为

$$T_n'+\kappa(\frac{n\pi}{l})^2T_n=f_n,$$

$$T_n(0)=\frac{2B(-1)^n}{n\pi}-\frac{2A}{n\pi}$$

其中f_n的表达式为

$$f_n(t)=\frac{2}{l}\int_0^l\left(A\alpha^2\kappa(1-\frac{x}{l})e^{-\alpha^2\kappa t}+B\beta^2\kappa\frac{x}{l}e^{-\beta^2\kappa t}\right)\sin\frac{n\pi x}{l}dx$$

$$= \frac{2}{n\pi}A\alpha^2\kappa e^{-\alpha^2\kappa t} - \frac{2}{n\pi}B(-1)^n\beta^2\kappa e^{-\beta^2\kappa t}$$

可解得T_n为

$$T_n(t) = e^{-\kappa(\frac{n\pi}{l})^2 t}(T_n(0) + \int_0^t f_n e^{\kappa(\frac{n\pi}{l})^2\tau}d\tau)$$

$$=e^{-\kappa(\frac{n\pi}{l})^2 t}[\frac{2B(-1)^n}{n\pi} - \frac{2A}{n\pi} + \frac{2\kappa}{n\pi}(\frac{A\alpha^2}{\kappa((\frac{n\pi}{l})^2-\alpha^2)}e^{((\frac{n\pi}{l})^2-\alpha^2)\kappa t} - \frac{B(-1)^n\beta^2}{\kappa((\frac{n\pi}{l})^2-\beta^2)}e^{((\frac{n\pi}{l})^2-\beta^2)\kappa t}$$
$$- \frac{A\alpha^2}{\kappa((\frac{n\pi}{l})^2-\alpha^2)} + \frac{B(-1)^n\beta^2}{\kappa((\frac{n\pi}{l})^2-\beta^2)})]$$

$$= \frac{2e^{-\kappa(\frac{n\pi}{l})^2 t}}{n\pi}(\frac{B(-1)^n(n\pi)^2}{(n\pi)^2-(\beta l)^2} - \frac{A(n\pi)^2}{(n\pi)^2-(\alpha l)^2}) + \frac{2}{n\pi}(\frac{A\alpha^2}{(\frac{n\pi}{l})^2-\alpha^2}e^{-\alpha^2\kappa t} - \frac{B(-1)^n\beta^2}{(\frac{n\pi}{l})^2-\beta^2}e^{-\beta^2\kappa t})$$

代回可得$u(x,t)$

$$u(x,t) = A(1-\frac{x}{l})e^{-\alpha^2\kappa t} + B\frac{x}{l}e^{-\beta^2\kappa t} + \sum_{n=1}^{\infty}T_n(t)\sin\frac{n\pi}{l}x$$

解法三（方程和边界条件同时齐次化）

设

$$u(x,t) = v(x,t) + w(x,t)$$

其中设$v(x,t)$形式为：

$$v(x,t) = Af(x)e^{-\alpha^2\kappa t} + Bg(x)e^{-\beta^2\kappa t}$$

且满足方程

$$\begin{cases}\dfrac{\partial v}{\partial t} - \kappa\dfrac{\partial^2 v}{\partial x^2} = 0,\\ v|_{x=0} = Ae^{-\alpha^2\kappa t},\ v|_{x=l} = Be^{-\beta^2\kappa t}\end{cases}$$

代入得：

$$\begin{cases}-A\alpha^2\kappa f(x)e^{-\alpha^2\kappa t} - B\beta^2\kappa g(x)e^{-\beta^2\kappa t} - \kappa\left(Af''(x)e^{-\alpha^2\kappa t} + Bg''(x)e^{-\beta^2\kappa t}\right) = 0,\\ f(0)=1,\ g(l)=1,\ f(l)=0,\ g(0)=0\end{cases}$$

整理得：

$$Ae^{-\alpha^2\kappa t}\left(f''(x)+\alpha^2 f(x)\right)+Be^{-\beta^2\kappa t}\left(g''(x)+\beta^2 g(x)\right)=0$$

设：

$$f''(x)+\alpha^2 f(x)=0,\ g''(x)+\beta^2 g(x)=0$$

结合上面的边界条件可解得：

$$f(x)=\frac{\sin\alpha(l-x)}{\sin\alpha l},\ g(x)=\frac{\sin\beta x}{\sin\beta l}$$

$$v(x,t)=A\frac{\sin\alpha(l-x)}{\sin\alpha l}e^{-\alpha^2\kappa t}+B\frac{\sin\beta x}{\sin\beta l}e^{-\beta^2\kappa t}$$

因此$w(x,t)$满足方程：

$$\begin{cases}\dfrac{\partial w}{\partial t}-\kappa\dfrac{\partial^2 w}{\partial x^2}=0,\\ w|_{x=0}=w|_{x=l}=0,\\ w|_{t=0}=-\left(A\frac{\sin\alpha(l-x)}{\sin\alpha l}+B\frac{\sin\beta x}{\sin\beta l}\right)\end{cases}$$

对$w(x,t)$进行本征函数展开：

$$w(x,t)=\sum_{n=1}^{\infty}C_n T_n(t)\sin\frac{n\pi}{l}x$$

代回方程可解得：

$$T_n(t)=e^{-\kappa\frac{n^2\pi^2}{l^2}t}$$

代入初始条件可得：

$$\sum_{n=1}^{\infty}C_n\sin\frac{n\pi x}{l}=A\frac{\sin\alpha(x-l)}{\sin\alpha l}-B\frac{\sin\beta x}{\sin\beta l}$$

$$C_n=\frac{2}{l}\int_0^l\sin\frac{n\pi x}{l}\left(A\frac{\sin\alpha(x-l)}{\sin\alpha l}-B\frac{\sin\beta x}{\sin\beta l}\right)dx$$

$$=\frac{2}{l}\left(\frac{A}{\sin\alpha l}\cdot\frac{-\frac{n\pi}{l}\sin\alpha l}{\left(\frac{n\pi}{l}\right)^2-\alpha^2}-\frac{B}{\sin\beta l}\cdot\frac{-\frac{n\pi}{l}\sin\beta l(-1)^n}{\left(\frac{n\pi}{l}\right)^2-\beta^2}\right)$$

$$=-\frac{2An\pi}{(n\pi)^2-(\alpha l)^2}+\frac{2Bn\pi(-1)^n}{(n\pi)^2-(\beta l)^2}$$

$$u(x,t)=A\frac{\sin\alpha(l-x)}{\sin\alpha l}e^{-\alpha^2\kappa t}+B\frac{\sin\beta x}{\sin\beta l}e^{-\beta^2\kappa t}+\sum_{n=1}^{\infty}(-\frac{2An\pi}{(n\pi)^2-(\alpha l)^2}+\frac{2Bn\pi(-1)^n}{(n\pi)^2-(\beta l)^2})\sin\frac{n\pi}{l}xe^{-\kappa\frac{n^2\pi^2}{l^2}t}$$

2 层状铀块临界厚度

考虑一维中子浓度满足的偏微分方程及其边界条件为

$$\begin{cases}\dfrac{\partial u}{\partial t}=D\dfrac{\partial^2 u}{\partial x^2}+\alpha u,\\ u|_{x=0}=u|_{x=l}=0\end{cases}$$

对$u(x,t)$进行分离变量

$$u(x,t)=T(t)X(x)$$

代回方程，设

$$\frac{T'}{T}=\frac{DX''+\alpha X}{X}=\lambda$$

可得分离变量后的方程和边界条件变为

$$\begin{cases}T'-\lambda T=0,\\ X''+\frac{\alpha-\lambda}{D}X=0,\\ X(0)=X(l)=0\end{cases}$$

可解得

$$\begin{cases}T_n(t)=A_ne^{\lambda_n t},\\ X_n(x)=\sin(\sqrt{\frac{\alpha-\lambda_n}{D}}x),\\ \sqrt{\frac{\alpha-\lambda_n}{D}}l=n\pi,\ n\in N*\end{cases}$$

解得

$$\lambda_n=\alpha-(\frac{n\pi}{l})^2D$$

因此

$$u(x,t)=\sum_{n=1}^{\infty}A_ne^{\lambda_n t}\sin\frac{n\pi x}{l}$$

临界厚度是指恰好使得层状铀块中的中子浓度不随时间增长也不衰减的厚度，此时中子产生率等于损失率，系统处于稳态临界状态。在上述方程中即指时间特征值$\lambda_n=0$。估算时

只考虑对$u(x,t)$数值影响最大的$n=1$项的时间特征值等于零，即

$$\alpha-(\frac{\pi}{l_{\max}})^2D=0$$

此时其余项的时间特征值

$$\lambda_n<0,\ n\geq 2$$

不会使得层状铀块中的中子浓度随时间增长，因此可解得临界厚度为

$$l_{\max}=\pi\sqrt{\frac{D}{\alpha}}$$

3 半无限带形区域电势分布

$$\begin{cases}u_{xx}+u_{yy}=0,\\ u|_{x=0}=u_{y=0}=0,\\ u|_{x=a}=u_0,\\ u|_{y\to\infty}\text{有界}\end{cases}$$

设

$$u(x,y)=v(x,y)+w(x,y)$$

其中

$$v(x,y)=\frac{u_0x}{a}$$

可得$w(x,y)$满足方程

$$\begin{cases}\dfrac{\partial^2w}{\partial x^2}+\dfrac{\partial^2w}{\partial y^2}=0,\\ w|_{x=0}=w|_{x=a}=0,\\ w|_{y=0}=-\frac{u_0x}{a},\ w|_{y\to\infty}\text{有界}\end{cases}$$

此时$w(x,y)$的本征函数为

$$X_n(x)=\sin\left(\frac{n\pi x}{l}\right)$$

$w(x,y)$可展开为

$$w(x,y)=\sum_{n=1}^{\infty}Y_n(y)\sin\frac{n\pi x}{a}$$

代回方程可得

$$Y_n'' - (\frac{n\pi}{a})^2 Y_n = 0$$

可解得

$$Y_n(y) = A_n e^{\frac{n\pi y}{a}} + B_n e^{-\frac{n\pi y}{a}}$$

根据边界条件可得

$$A_n = 0,$$

$$\sum_{n=1}^{\infty} B_n \sin\frac{n\pi x}{a} = -\frac{u_0 x}{a}$$

解得

$$B_n = -\frac{2u_0}{a^2}\int_0^a x\sin\frac{n\pi x}{a}dx = \frac{2u_0(-1)^n}{n\pi}$$

因此

$$u(x,y) = \frac{u_0 x}{a} + \sum_{n=1}^{\infty}\frac{2u_0(-1)^n}{n\pi}e^{-\frac{n\pi y}{a}}\sin\frac{n\pi x}{a}$$

本题也可采用展开法求解（略）。

4 弦振动问题

解法1：展开法（直接按本征函数展开）

(1) 齐次边界条件下的方程

$$\begin{cases}\dfrac{\partial^2 u}{\partial t^2} - a^2\dfrac{\partial^2 u}{\partial x^2} = 0,\\ u|_{x=0} = u|_{x=l} = 0\end{cases}$$

其本征函数为

$$X_n(x) = \sin\left(\frac{n\pi x}{l}\right)$$

原方程的解用本征函数展开

$$u(x,t) = \sum_{n=1}^{\infty} T_n(t)X_n(x) = \sum_{n=1}^{\infty} T_n(t)\sin\left(\frac{n\pi x}{l}\right), \quad t>0$$

对原方程作积分

$$\frac{2}{l}\int_0^l \left(\frac{\partial^2 u}{\partial t^2} - a^2\frac{\partial^2 u}{\partial x^2}\right) X_n(x)dx = \frac{2}{l}\int_0^l f(x)\cos\left(\frac{2\pi a}{l}t\right) X_n(x)dx$$

其中第一项中

$$\frac{2}{l}\int_0^l u(x,t)X_n(x)dx = \frac{2}{l}\int_0^l u(x,t)\sin\left(\frac{n\pi x}{l}\right)dx = T_n(t)$$

第二项中

$$a^2\frac{1}{l}\int_0^l \frac{\partial^2 u}{\partial x^2}X_n(x)dx = a^2\frac{1}{l}\left[\int_0^l X_n''(x)u(x,t)dx + (u'(x,t)X_n(x) - u(x,t)X_n'(x))\,|_0^l\right]$$

$$= -\left(\frac{n\pi a}{l}\right)^2 T_n(t) - (-1)^n\frac{2n\pi a^2}{l^2}A\cos\left(\frac{2\pi a}{l}t\right)$$

周期性外力项中

$$\frac{2}{l}\int_0^l f(x)X_n(x)dx = \frac{2}{l}\int_c^d F\sin\left(\frac{n\pi x}{l}\right)dx = \frac{2}{n\pi}F\left[\cos\left(\frac{n\pi c}{l}\right) - \cos\left(\frac{n\pi d}{l}\right)\right]$$

由上述三式可得 $T_n(t)$ 满足的方程为

$$T_n''(t) + \left(\frac{n\pi a}{l}\right)^2 T_n(t) = A_n\cos\left(\frac{2\pi a}{l}t\right), \quad t > 0$$

其中

$$A_n = \frac{2}{n\pi}F\left[\cos\left(\frac{n\pi c}{l}\right) - \cos\left(\frac{n\pi d}{l}\right)\right] - (-1)^n\frac{2n\pi a^2}{l^2}A.$$

初始条件

$$T_n(0) = 0, \quad T_n'(0) = 0$$

可以采用齐次化原理解出 $T_n(t)$。首先写下齐次化原理，

$$T_n(t) = \int_0^t s(t;\tau)d\tau, \quad t > 0$$

然后求出 $s(t;\tau)$，积分给出 $n \neq 2$和 $n = 2$的解。这里 $s(t,\tau)$ 满足的齐次方程

$$\frac{d^2 s(t;\tau)}{dt^2} + \left(\frac{n\pi a}{l}\right)^2 s(t;\tau) = 0$$

以及初始条件

$$s|_{t=\tau}=0,\quad \frac{ds}{dt}|_{t=\tau}=A_n\cos\left(\frac{2\pi a}{l}\tau\right)$$

由此得到

$$s(t;\tau)=A_n\frac{l}{n\pi a}\sin\frac{n\pi a}{l}(t-\tau)\cos\frac{2\pi a}{l}\tau$$

对 τ 积分得

$$T_n(t)=\begin{cases}\dfrac{1}{n^2-4}\left(\dfrac{l}{n\pi a}\right)^2A_n\left[\cos\left(\dfrac{2\pi a}{l}t\right)-\cos\left(\dfrac{n\pi a}{l}t\right)\right], & n\neq 2,\\ \dfrac{l}{4\pi a}A_2t\sin\left(\dfrac{2\pi a}{l}t\right), & n=2.\end{cases}$$

代入

$$u(x,t)=\sum_{n=1}^{\infty}T_n(t)X_n(x)=\sum_{n=1}^{\infty}T_n(t)\sin\left(\frac{n\pi x}{l}\right),\quad t>0$$

(2) 共振来自 $n=2$ 项，消除共振要求 $A_2=0$，代入 A_n式中可得

$$A=\frac{l^2}{4\pi^2a^2}F\left[\cos\left(\frac{2\pi c}{l}\right)-\cos\left(\frac{2\pi d}{l}\right)\right]$$

解法2：减去法（边界条件齐次化）

(1)设 $u(x,t)=w(x,t)+p(x,t)$，其中设齐次化函数

$$p(x,t)=A\frac{x}{l}\cos\left(\frac{2\pi a}{l}t\right)$$

满足边界条件。这样得到 $w(x,t)$ 满足

$$\begin{cases}\dfrac{\partial^2w}{\partial t^2}-a^2\dfrac{\partial^2w}{\partial x^2}=\left[f(x)+\dfrac{4\pi^2a^2}{l^2}A\dfrac{x}{l}\right]\cos\left(\dfrac{2\pi a}{l}t\right),\\ w|_{x=0}=w|_{x=l}=0,\\ w|_{t=0}=-A\dfrac{x}{l},\quad \dfrac{\partial w}{\partial t}|_{t=0}=0.\end{cases}$$

设 $w(x,t)=w_1(x,t)+w_2(x,t)$，其中 $w_1(x,t)$ 承担初始条件，$w_2(x,t)$ 承担非齐次项。w_1

满足方程

$$\begin{cases} \dfrac{\partial^2 w_1}{\partial t^2} - a^2 \dfrac{\partial^2 w_1}{\partial x^2} = 0, \\ w_1|_{x=0} = w_1|_{x=l} = 0, \\ w_1|_{t=0} = -A\dfrac{x}{l}, \quad \dfrac{\partial w_1}{\partial t}|_{t=0} = 0. \end{cases}$$

一般解为

$$w_1(x,t) = \sum_{n=1}^{\infty} \left[A_n \cos\left(\frac{n\pi a}{l}t\right) + B_n \sin\left(\frac{n\pi a}{l}t\right) \right] \sin\left(\frac{n\pi}{l}x\right)$$

由初始条件

$$w_1|_{t=0} = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi}{l}x\right) = -A\frac{x}{l},$$

$$\frac{\partial w_1}{\partial t}|_{t=0} = \sum_{n=1}^{\infty} n\pi a B_n \sin\left(\frac{n\pi}{l}x\right) = 0,$$

得到

$$A_n = -A\frac{2}{l}\int_0^l \frac{x}{l}\sin\left(\frac{n\pi}{l}x\right)dx = (-1)^n \frac{2}{n\pi}A, \quad B_n = 0$$

这样得到

$$w_1(x,t) = \sum_{n=1}^{\infty} (-1)^n \frac{2}{n\pi} A \cos\left(\frac{n\pi a}{l}t\right) \sin\left(\frac{n\pi}{l}x\right).$$

$w_2(x,t)$ 满足非齐次方程组

$$\begin{cases} \dfrac{\partial^2 w_2}{\partial t^2} - a^2 \dfrac{\partial^2 w_2}{\partial x^2} = \left[f(x) + \dfrac{4\pi^2 a^2}{l^2} A \dfrac{x}{l} \right] \cos\left(\dfrac{2\pi a}{l}t\right), \\ w_2|_{x=0} = w_2|_{x=l} = 0, \\ w_2|_{t=0} = 0, \quad \dfrac{\partial w_2}{\partial t}|_{t=0} = 0. \end{cases}$$

用齐次化原理求解 $w_2(x,t)$：

$$w_2(x,t) = \int_0^t s(x,t;\tau)d\tau,$$

$s(x,t;\tau)$ 满足

$$\begin{cases} \dfrac{\partial^2 s}{\partial t^2} - a^2 \dfrac{\partial^2 s}{\partial x^2} = 0, \\ s|_{x=0} = s|_{x=l} = 0, \\ s|_{t=\tau} = 0, \quad \dfrac{\partial s}{\partial t}|_{t=\tau} = \left[f(x) + \dfrac{4\pi^2 a^2}{l^2} A \dfrac{x}{l} \right] \cos\left(\dfrac{2\pi a}{l} \tau \right). \end{cases}$$

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一般解为

$$s(x,t;\tau) = \sum_{n=1}^{\infty} \left[C_n \cos\left(\frac{n\pi a}{l}(t-\tau) \right) + D_n \sin\left(\frac{n\pi a}{l}(t-\tau) \right) \right] \sin\left(\frac{n\pi x}{l} \right),$$

由初始条件

$$s|_{t=\tau} = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi}{l} x \right) = 0,$$

$$\frac{\partial s}{\partial t}|_{t=\tau} = \sum_{n=1}^{\infty} \frac{n\pi a}{l} D_n \sin\left(\frac{n\pi}{l} x \right) = \left[f(x) + \frac{4\pi^2 a^2}{l^2} A \frac{x}{l} \right] \cos\left(\frac{2\pi a}{l} \tau \right),$$

得到

$$C_n = 0, \quad D_n = \frac{l}{n\pi a} M_n \cos\left(\frac{2\pi a}{l} \tau \right),$$

其中

$$M_n = \frac{2}{l} \int_0^l \left[f(x) + \frac{4\pi^2 a^2}{l^2} A \frac{x}{l} \right] \sin\left(\frac{n\pi}{l} x \right) dx$$

$$= \frac{2}{n\pi} F \left[\cos\left(\frac{n\pi c}{l} \right) - \cos\left(\frac{n\pi d}{l} \right) \right] - (-1)^n \frac{8\pi a^2}{n l^2} A.$$

则

$$s(x,t;\tau) = \sum_{n=1}^{\infty} \frac{l}{n\pi a} M_n \sin\left(\frac{n\pi}{l} x \right) \cos\left(\frac{2\pi a}{l} \tau \right) \sin\left(\frac{n\pi}{l}(t-\tau) \right),$$

对 τ 作积分得到

$$w_2(x,t) = \int_0^t s(x,t;\tau) d\tau$$

$$= \sum_{n=1}^{\infty} \frac{l^2}{(n^2-4)\pi^2 a^2} M_n \sin\left(\frac{n\pi}{l} x \right) \left[\cos\left(\frac{2\pi a}{l} t \right) - \cos\left(\frac{n\pi a}{l} t \right) \right]$$

$$+ \frac{l}{4\pi a} M_2 t \sin\left(\frac{2\pi a}{l} t \right) \sin\left(\frac{2\pi}{l} x \right).$$

综上，有

$$u(x,t) = p(x,t) + w_1(x,t) + w_2(x,t)$$

$$= A\frac{x}{l}\cos\left(\frac{2\pi a}{l}t\right) + \sum_{n=1}^{\infty}(-1)^n\frac{2}{n\pi}A\cos\left(\frac{n\pi a}{l}t\right)\sin\left(\frac{n\pi}{l}x\right)$$

$$+\sum_{n=1}^{\infty}\frac{l^2}{(n^2-4)\pi^2a^2}M_n\sin\left(\frac{n\pi}{l}x\right)\left[\cos\left(\frac{2\pi a}{l}t\right) - \cos\left(\frac{n\pi a}{l}t\right)\right]$$

$$+\frac{l}{4\pi a}M_2 t\sin\left(\frac{2\pi a}{l}t\right)\sin\left(\frac{2\pi}{l}x\right).$$

(2)共振来自 $n=2$ 项，消除共振要求 $M_2=0$，代入M_n得到

$$A = \frac{l^2}{4\pi^2a^2}F\left[\cos\left(\frac{2\pi c}{l}\right) - \cos\left(\frac{2\pi d}{l}\right)\right].$$

注：也可用通解加特解的方法解方程$T_n''(t) + \left(\frac{n\pi a}{l}\right)^2 T_n(t) = A_n\cos\left(\frac{2\pi a}{l}t\right), \quad t>0$
设

$$T_n = T_n^{(1)}(t) + T_n^{(2)}(t),$$

其中 $T_n^{(1)}(t)$ 为通解，$T_n^{(2)}(t)$ 为特解。特解 $T_n^{(2)}(t)$ 满足T_n方程，设为

$$T_n^{(2)}(t) = B_n\cos\left(\frac{2\pi a}{l}t\right),$$

代入T_n方程，得到

$$B_n = \frac{1}{n^2-4}\left(\frac{l}{\pi a}\right)^2 A_n, \quad n\neq 2.$$

$n=2$ 时处于共振情形，上面特解不成立，特解设为

$$T_n^{(2)}(t) = B_2 t\sin\left(\frac{2\pi a}{l}t\right),$$

代入T_n方程，得到

$$B_2 = \frac{l}{4\pi a}A_2.$$

这样得到特解

$$T_n^{(2)}(t)=\begin{cases}\dfrac{1}{n^2-4}\left(\dfrac{l}{\pi a}\right)^2 A_n\cos\left(\dfrac{2\pi a}{l}t\right), & n\neq 2,\\ \dfrac{l}{4\pi a}A_2 t\sin\left(\dfrac{2\pi a}{l}t\right), & n=2.\end{cases}$$

通解 $T_n^{(1)}(t)$ 满足T_n方程对应的齐次方程，设为

$$T_n^{(1)}(t)=C_n\cos\left(\frac{n\pi a}{l}t\right)+D_n\sin\left(\frac{n\pi a}{l}t\right).$$

由特解$T_n^{(2)}(t)$，得到 $T_n^{(1)}(t)$ 的初始条件

$$T_n^{(1)}(0)=\begin{cases}-\dfrac{1}{n^2-4}\left(\dfrac{l}{\pi a}\right)^2 A_n, & n\neq 2,\\ 0, & n=2,\end{cases}$$

代入$T_n^{(1)}(t)$，得到

$$C_n=\begin{cases}-\dfrac{1}{n^2-4}\left(\dfrac{l}{\pi a}\right)^2 A_n, & n\neq 2,\\ 0, & n=2,\end{cases}$$

由此

$$T_n^{(1)}(t)=\begin{cases}-\dfrac{1}{n^2-4}\left(\dfrac{l}{\pi a}\right)^2 A_n\cos\left(\dfrac{n\pi a}{l}t\right), & n\neq 2,\\ 0, & n=2.\end{cases}$$

这样得到

$$T_n(t)=T_n^{(1)}(t)+T_n^{(2)}(t)=\begin{cases}\dfrac{1}{n^2-4}\left(\dfrac{l}{\pi a}\right)^2 A_n\left[\cos\left(\dfrac{2\pi a}{l}t\right)-\cos\left(\dfrac{n\pi a}{l}t\right)\right], & n\neq 2,\\ \dfrac{l}{4\pi a}A_2 t\sin\left(\dfrac{2\pi a}{l}t\right), & n=2.\end{cases}$$

代入

$$u(x,t)=\sum_{n=1}^{\infty}T_n(t)X_n(x)=\sum_{n=1}^{\infty}T_n(t)\sin\left(\frac{n\pi x}{l}\right),\quad t>0.$$